



Original Article

An inter-comparison between ENDF/B-VIII.0-NECP-Atlas and ENDF/B-VIII.0-NJOY results for criticality safety benchmarks and benchmarks on the reactivity temperature coefficient

Ouahie Kabach^{a,*}, Abdelouahed Chetaine^a, Abdelfettah Benchrif^b, Hamid Amsil^{a,b}^a Mohammed V University, Faculty of Science, Nuclear Reactor and Nuclear Security Group Energy Centre, Physics Department, 4 Avenue Ibn Battouta B.P. 1014 RP, Rabat, 10000, Morocco^b National Centre for Nuclear Energy, Sciences and Technology (CNESTEN), Morocco

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ABSTRACT

Since the nuclear data forms a vital component in reactor physics computations, the nuclear community needs processing codes as tools for translating the Evaluated Nuclear Data Files (ENDF) to simulate nuclear-related problems such as an ACE format that is used for MCNP. Errors, inaccuracies or discrepancies in library processing may lead to a calculation that disagrees with the experimentally measured benchmark. This paper provides an overview of the processing and preparation of ENDF/B-VIII.0 incident neutron data with NECP-Atlas and NJOY codes for implementation in the MCNP code. The resulting libraries are statistically inter-compared and tested by conducting benchmark calculations, as the mutual-comparison is a source of strong feedback for further improvements in processing procedures. The database of the benchmark experiments is based on a selection taken from the International Handbook of Evaluated Criticality Safety Benchmark Experiments (ICSBEP handbook) and those proposed by Russell D. Mosteller. In general, there is quite good agreement between the NECP-Atlas1.2 and NJOY21^(1.0.0.json) results with no substantial differences, if the correct input parameters are used.

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1. Introduction

Nuclear evaluations are physics representations of the data derived from experimental data and theoretical predictions encoded in a unified computer-readable format called Evaluated Nuclear Data File (ENDF) released by the U.S. Cross Section Evaluation Working Group (CSEWG) [1]. ENDF files cannot be used directly, they need to be converted into suitable forms for applications, such as calculations of particle transport using multi-group, pointwise for deterministic and Monte Carlo techniques. As a result, cross-section processing codes such as NJOY [2], NECP-Atlas [3], PREPRO [4] and AMPX [5] must be used to translate and manipulate the data from the single universal [6] ENDF format to a variety of formats accessible via particle transport codes.

Another point to be taken into account is that the use of different sets of processing codes in the analysis of a benchmark

problem is both an essential feature and a source of strong feedback for further improvements in data through mutual comparison of results. In fact, the differences observed in the results obtained are not only due to cross-sections but also to the calculation methodologies used by these codes (i.e. interpolation, cross-section tolerances, probability tables, etc.) [7,8]. Other relevant information is the type of verification tests conducted and the types of benchmarks used to validate the libraries. This can be done by checking the processed libraries against benchmarks that are recognized internationally to ensure that the libraries are capable of providing accurate and reliable results. Further, the Monte Carlo approach is the most suitable for directly validating nuclear data libraries since processed pointwise cross-section data in terms of ACE format (A Compact ENDF) can be used directly without the need of a preceding resonance treatment [7].

In the current work, we describe how the neutron data file is processed with NECP-Atlas and NJOY into ACE libraries. Besides, numerical tests, along with statistical tests, are carried out to provide a code-to-code comparison, and to obtain a more comprehensive impression and understanding of the criticality quality of the processed neutron libraries. The results presented in this paper

* Corresponding author.

E-mail addresses: ouadie.kabach10@gmail.com, ouadie_kabach@um5.ac.ma (O. Kabach).

are based on a set of benchmark calculations, including 120 criticality safety benchmarks, and a set of numerical benchmarks for the Doppler coefficient of reactivity. The criticality benchmarks are taken from the ICSBEP handbook [13], whereas the Doppler benchmarks are given by Refs. [14–16]. The Monte Carlo N-Particle transport code (MCNPX) was used for the present computations [17].

2. Description of nuclear data and codes used

2.1. The ENDF/B-VIII.0 neutron library

The ENDF/B nuclear data library is a collection of evaluated nuclear data files released by the Cross Section Evaluation Working Group (CSEWG) [7]. The ENDF/B library contains in its diverse releases over the past years neutron and proton interaction data, thermal scattering law data, decay data, photonuclear and electron data and spontaneous fission yields. The following work addresses only the treatment of ENDF/B-VIII.0 neutron data, which is the newest ENDF/B library released on February 2, 2018 [9,10,18]. The ENDF/B-VIII.0 neutron data library provides cross-section data for 557 nuclides and thermal scattering libraries for 33 different compounds in ENDF-6 files.

2.2. Descriptions of processing codes

The nuclear data processing codes used for this work are NECP-Atlas and NJOY. Both codes can be used to convert the ENDF-6 format into pointwise and multigroup cross-sections, but in this work, they are being used to produce pointwise cross-sections data in terms of ACE libraries. In order to eliminate any differences in the processed libraries, we decided to generate libraries with the same basic set of input parameters for all available nuclear data (including the $s(\alpha, \beta)$ thermal scattering data treatments) as given in Table 1. Additionally, the temperatures treated are 293.6, 600, and 900 K.

Data processing using NECP-Atlas v1.2

NECP-Atlas is a modular code system developed at Xi'an Jiaotong University in China to produce accurate cross-sections for different application aspects. NECP-Atlas is written with the object-oriented method in standard FORTRAN 2008, and various processing functions are implemented in different modules. The code can easily process ENDF-6 format evaluations and convert them into ACE formats that are applicable for different applications such as MCNP [3,11,12]. The processing sequence of the modules implemented into NECP-Atlas is given in Fig. 1.

Data processing using NJOY21

The NJOY21 Nuclear Data Processing System is a software system used to read evaluated nuclear data information stored in ENDF-6 to produce different types of formats useful for criticality or shielding applications [19]. The modules used in this processing are MODER, RECONR, BROADR, HEATR, THERMR, GASPR, PURR and ACER for generating the ACE formatted library [2,7]. The processing sequence is presented in Fig. 1.

Table 1
Processing options used in a typical input file for NECP-Atlas and NJOY codes.

Reconstruction tolerance:	0.1%
Resonance integral check tolerance	0.3%
Reconstruction temperature:	0.0 K
Thinning tolerance	0.1%
Maximum energy for Doppler broadening	2 MeV

2.3. Monte Carlo simulations

All Monte Carlo simulations described in the following have been carried out by resorting to MCNPX computer code [17]. MCNP, the general-purpose Monte Carlo radiation transport code developed at Los Alamos National Laboratory and dedicated to criticality safety, reactor physics with depletion, and shielding [20]. The code offers criticality and fixed source simulation mode and several options for the k_{eff} value accompanied by an associated uncertainty $|\Delta k_{\text{eff}}|$. For this study, the combined average of absorption, collision, and track-length estimator is quoted as the k_{eff} value. Each of the cases employed 600 generations of 50,000 neutron histories each and the results from the first 100 generations were discarded for the determination of the k_{eff} . Consequently, the results reported herein are based on 25,000,000 active neutrons histories for each case. Statistical uncertainties with 1σ reliability of k_{eff} values are less than 0.00022 in all cases.

3. Methodology and problem specification

3.1. Selected benchmarks

In the nuclear context, the term benchmark is widely used to compare standard experiments with calculations. Benchmark can be an experiment performed for certain fuel composition and arrangement or a given geometry configuration, which is considered as a reference for efficiency and convergence of the results. Benchmark also serves as a tool to validate calculation techniques, nuclear data libraries or computer codes, including code-to-code comparison. The following subsections include descriptions of the benchmarks used.

Criticality safety benchmarks

The analyzed benchmarks were selected to obtain a set of problems that would address different criteria, such as fissile isotope, energy regions (i.e. thermal, intermediate, and fast regions), reflectors, and structural materials. The geometry and material specifications for the 120 selected benchmarks were taken from the ICSBEP handbook [13]. In this series, there are five sub-categories: critical assemblies with ^{233}U , Low enriched, natural ^{235}U (LEU), intermediate enriched ^{235}U (IEU), highly enriched ^{235}U (HEU), ^{239}Pu or mixed metal assemblies. In addition, different moderator and reflector assemblies (e.g. Graphite, Beryllium, Unreflected, etc.) characterize each category. For more information about the criticality safety benchmark, we refer to Refs. [21,22].

Benchmarks on the reactivity temperature coefficient

The Doppler defect benchmarks have been proposed in 1991 to assess the Doppler coefficient of reactivity feedback for 2D cells [14]. In fact, they have relatively simple geometries and they hold all of the important contributors to Doppler reactivity feedback. The benchmark data are based on previous benchmark studies and they include seven different fuel enrichments of a typical pressurized water reactor LWR- UO_2 lattice, five different mixed oxides (MOX) concentrations of Reactor-grade fuel and four different MOX concentrations of Weapons-grade fuel. The isotopic concentrations of fuels, cladding, and the moderator are taken from Refs. [15,16]. Benchmark configuration corresponds to infinite corresponding pairs of cells consisting of fuel, void, cladding and borated water. At the hot zero power (HZIP); the temperature for fuel, cladding, and the borated moderator is uniform 600 K. At the hot full power (HFP), the fuel temperature is 900 K, while the temperature of everything else remains at 600 K. Moreover, the benchmark took into account the variation in fuel pellet diameter as a result of temperature difference. The Doppler defect (D_C) can be simply evaluated from the reactivity difference between HFP and HZIP conditions which is mathematically presented in Eq. (1) [14]. The pin cell dimensions are given in Table 2.

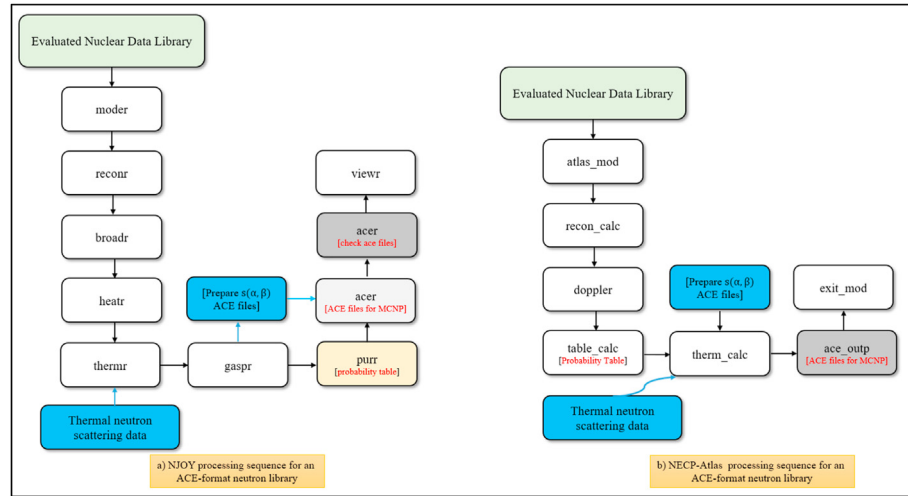


Fig. 1. NJOY and NECP-Atlas processing sequences for an ACE-format neutron library.

Table 2
Configuration of PWR unit cell.

Parameter	HZP	HFP
Unit cell pitch, (cm)	1.26678	
Outer radius of fuel, (cm)	0.39398	0.39433
Inner radius of cladding, (cm)	0.40226	0.40226
Outer radius of cladding, (cm)	0.45972	0.45972
Cladding material	Zirconium	
Moderator material	Water with 1400 ppm of Boron	

$$D_C = \frac{\Delta \rho}{\Delta T} = \frac{\rho(T_{HFP}) - \rho(T_{HZP})}{T_{HFP} - T_{HZP}} \quad (1)$$

3.2. Comparison methods

In this section, we explain the methods used to measure the values reported in this paper. The comparison between the codes is reported using the calculated k_{eff} normalized to the benchmark ($\frac{C}{E}$)

and its estimated uncertainty ($\sigma_{(\frac{C}{E})}$) together with the relative

difference from the unity value (Δk_{eff}) [21,23]. For the criticality benchmarks the expected k_{eff} value for were taken from the ICSBEP and are presented with the results. The $\frac{C}{E}$ is calculated as shown in Eq. (2), and its uncertainty is determined as presented in Eq. (3). In Eq. (4), ΔX indicates the relative difference, and X is the analyzed parameter (k_{eff} and D_C).

$$\frac{C}{E} = \frac{k_{eff}^{calculated}}{k_{eff}^{expected}} \quad (2)$$

$$\sigma_{(\frac{C}{E})} = \frac{C}{E} \left[\sqrt{\left(\frac{\sigma_{expected}}{k_{eff}^{expected}} \right)^2 + \left(\frac{\sigma_{calculated}}{k_{eff}^{calculated}} \right)^2} \right] \quad (3)$$

$$\Delta X = \left(\frac{X^{calculated}}{X^{expected}} - 1 \right) \quad (4)$$

In addition to the $\frac{C}{E}$ values, several statistical metrics are compared to determine the accuracy between the calculations and which result shows the better fit to the benchmark values. These metrics include:

- The uncertainty of the expected results ($\sigma_{expected}$), and the uncertainty of the calculated results ($\sigma_{calculated}$) are combined statistically to give a combined statistical uncertainty (σ_s) Eq. (5). The combined uncertainty is used to determine if the results of both processing codes are statistically identical. We can adopt that the results are identical when the relative difference is within a $\pm 3 \sigma_s$ interval (99.6% of confidence) [23].

$$\sigma_s = \sqrt{(\sigma_{expected})^2 + (\sigma_{calculated})^2} \quad (5)$$

- The χ^2 (chi-square) goodness-of-fit test, and the average difference between calculated and expected k_{eff} $\langle |\Delta| \rangle$ [24], Eqs. (6) and (7).

$$\chi^2 = \sum \frac{((k_{calc} - k_{exp}) / \sigma_{expected})^2}{n} \quad (6)$$

$$\langle |\Delta| \rangle = \sum \frac{|k_{calc} - k_{exp}|}{n} \quad (7)$$

The statistical analysis is also conducted by determining the weighted mean multiplication factor ($\overline{k_{eff}}$) Eq. (8) and the bias uncertainty (S_p) Eq. (9). The two approaches involve determining the variance about the mean (s^2) Eq. (10) and the total average uncertainty ($\overline{\sigma_s}^2$) Eq. (11) [25–27]. Where n is the number of critical experiments used in the validation and $k_{eff,i}$ is the i th value of the multiplication factor.

$$\overline{k_{\text{eff}}} = \frac{\sum_i \frac{1}{\sigma_{s,i}^2} (k_{\text{eff},i})}{\sum_i \frac{1}{\sigma_{s,i}^2}} \quad (8)$$

$$S_p = \sqrt{s^2 + \overline{\sigma_s}^2} \quad (9)$$

$$s^2 = \frac{\frac{1}{(n-1)} \sum_i \frac{1}{\sigma_{s,i}^2} (k_{\text{eff},i} - \overline{k_{\text{eff}}})^2}{\frac{1}{n} \sum_i \frac{1}{\sigma_{s,i}^2}} \quad (10)$$

$$\overline{\sigma_s}^2 = \frac{n}{\sum_i \frac{1}{\sigma_{s,i}^2}} \quad (11)$$

4. Results

4.1. Results of criticality calculations

This section presents the analysis of the 120 critical benchmarks results. This takes account of the MCNP results of the calculations (i.e. NECP-Atlas and NJOY results compared to the benchmark values) and a detailed statistical inter-comparison between NECP-Atlas and NJOY by benchmark category. Fig. 2 to Fig. 6 show the results of the MCNP calculations based on ENDF/B-VIII.0 data library expressed as normalized to the benchmark value ($\frac{C}{C_E} \pm \sigma \left(\frac{C}{C_E} \right)$) along with Δk_{eff} . The benchmark uncertainty is also plotted (marked in grey) to visualize the tendencies in the results. Besides, for the majority of studied cases converged to a Monte Carlo stochastic uncertainty between ± 7 pcm and ± 22 pcm in k_{eff} .

It can be marked from Fig. 2 to Fig. 6 that most of the computed k_{eff} values are inside of experimental error bands (95% of cases are within 1σ level), within Δk_{eff} 's around 1000 pcm except for five-benchmark problems. Both uranyl-fluoride solutions (i.e. 'leu-sol-therm-001' and 'u233-sol-inter-001-1') and plutonium-nitrate solutions (i.e. 'pu-sol-therm-009-3a' and 'pu-sol-therm-011-181') categories show poor agreement with the expected results since the obtained Δk_{eff} is around 1.5% on an absolute basis. The largest deviations from unity also occur for the IEU (37.5 wt%) reflected by uranium and steel (mix-met-fast-008) and are more than 2% Δk_{eff} on an absolute basis. However, it is worth mentioning that this pattern also has been reported for earlier ENDF/B releases [28–32]].

4.2. Statistical analysis

Using the data in Fig. 2 to Fig. 6, this subsection provides a statistical evaluation between NECP Atlas and NJOY for each case in each category of benchmarks. Fig. 7 to Fig. 11 show comparisons between NECP-Atlas and NJOY based $\frac{\Delta k_{\text{eff}}}{k_{\text{eff}}}$ and $\frac{\Delta k_{\text{eff}}}{\sigma_s}$ metrics for each of the benchmark systems regrouped by benchmarks category. Tables 3 and 4 show the validation metrics as calculated by Eqs. (4 to 11) for the whole category. From the plotted results, it is observed that majority of the cases show good agreement between NECP-Atlas and NJOY with $|\Delta k_{\text{eff}}|$ values within 0.02% of each other. Few cases exhibit differences between 0.04% and 0.05%. However, more than 54% of k_{eff} values based on NECP-Atlas were higher than NJOY values. Concerning the obtained uncertainties, 93% of the cases are within 3σ , 77% of the cases are within 2σ and 47% of the cases are within 1σ . Moreover, the LEU systems show the most significant difference NECP-Atlas and NJOY results (93% of cases). Yet, based on the virtually identical values of χ^2 , $\langle |\Delta| \rangle$ and $\overline{k_{\text{eff}}} \pm S_p$, show that there are minor differences between NECP-Atlas and NJOY processing codes.

4.3. Results for the Doppler benchmark

The multiplication factors were calculated at HZP and HFP for various types of fuel. The Doppler coefficient was calculated according to Eq. (1) and ΔD_c using Eq. (4). We chose to use the results obtained from the ENDF/B-VII.0 library as a reference. The results are shown in Fig. 12 to Fig. 14, the relative error in the Doppler coefficient is also plotted in the above-mentioned figures.

When we compare the results from ENDF/B-VII.0 and ENDF/B-VIII.0 cross-sections, we find that there is no straight rule between the calculated k-infinity values (i.e. ENDF/B-VIII.0 values sometimes higher sometimes lower). Besides, the maximum difference in k-infinity between the two libraries is up to 0.4% on an absolute basis. Conversely, there is a difference in the Doppler coefficient since ΔD_c is around 4%, 3% and 3% on an absolute basis, for UO₂, reactor-recycled and weapons-grade MOX fuel category, respectively, but they are mainly within the $\pm 3\sigma_s$ interval, near the ENDF/B-VII.0 values. Also, the maximum ΔD_c occurs for the Weapons-Grade MOX fuel case 1 (9% on an absolute basis).

The comparison between ENDF/B-VIII.0-NECP-Atlas and ENDF/B-VIII.0-NJOY indicates that the NECP-Atlas-based k-infinity values are slightly below the NJOY values (60% of the total values) but differences are below 0.04% on an absolute basis. On the other hand, more than 70% of the D_c values calculated with ENDF/B-VIII.0-NECP-Atlas are higher than the values given by ENDF/B-VIII.0-NJOY, that may be due to the use of different cross-section

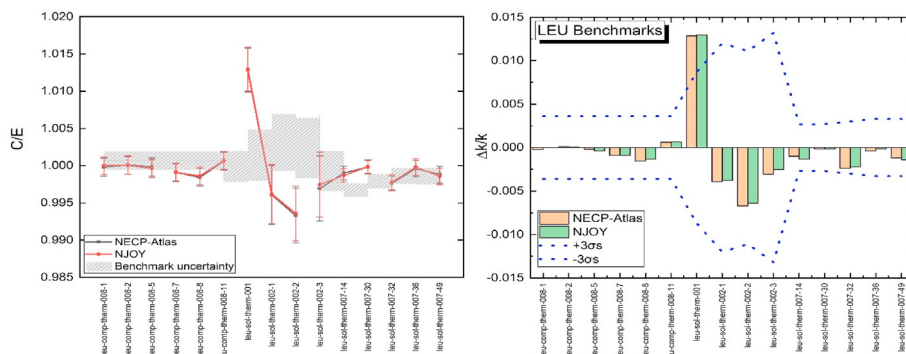


Fig. 2. $\left(\frac{C}{C_E}\right)$ and Δk_{eff} results for LEU benchmarks.

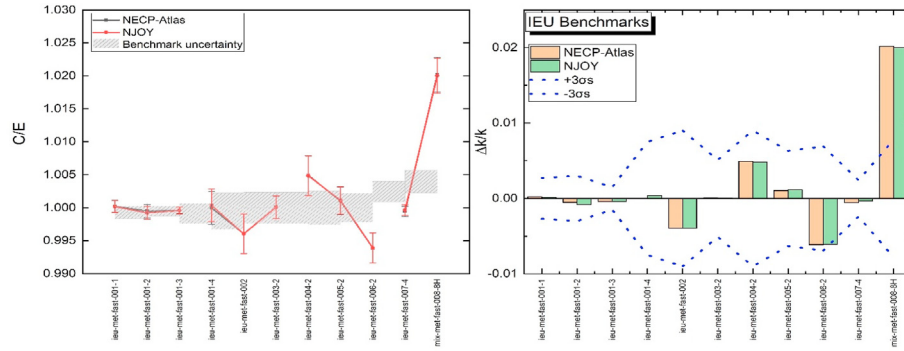


Fig. 3. $(\bar{\epsilon})$ and Δk_{eff} results for IEU benchmarks.

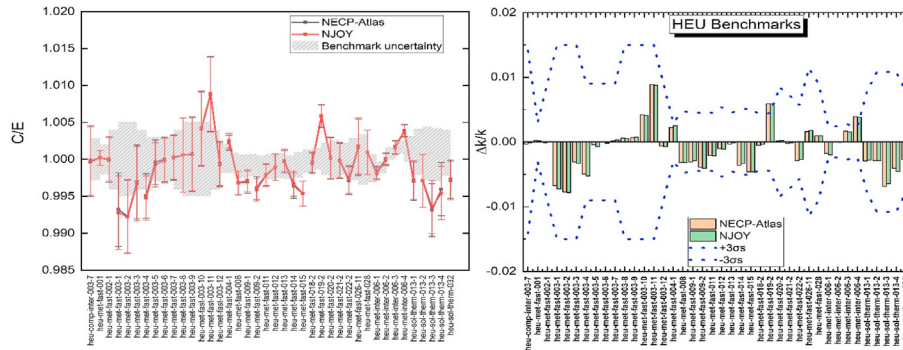


Fig. 4. $(\bar{\epsilon})$ and Δk_{eff} results for HEU benchmarks.

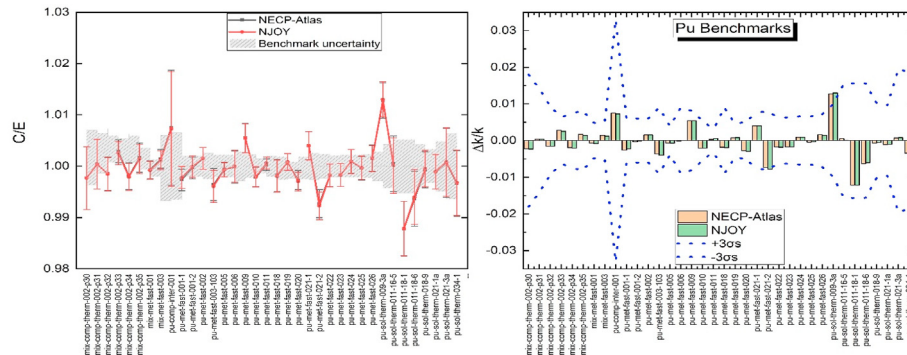


Fig. 5. $(\bar{\epsilon})$ and Δk_{eff} results for Pu benchmarks.

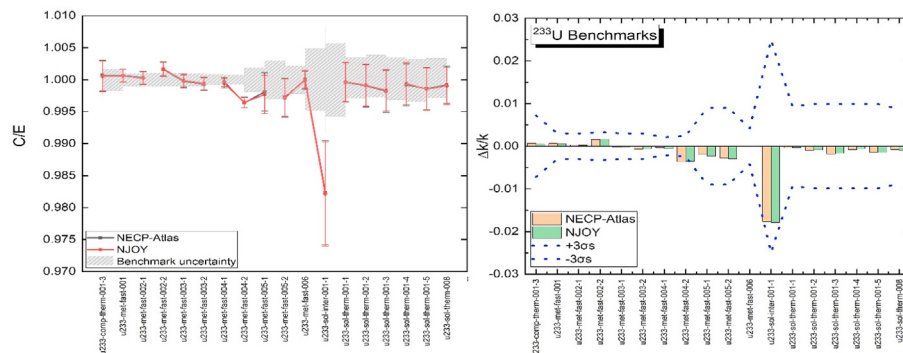


Fig. 6. $(\bar{\epsilon})$ and Δk_{eff} results for ^{233}U benchmarks.

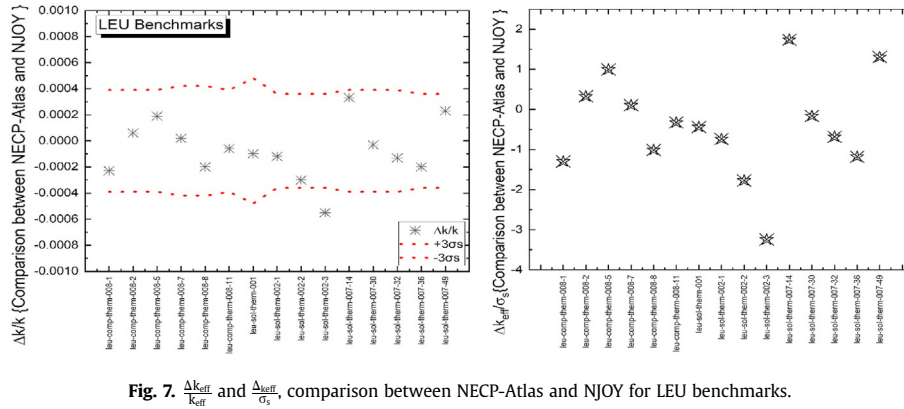


Fig. 7. $\frac{\Delta k_{eff}}{k_{eff}}$ and $\frac{\Delta k_{eff}}{\sigma_s}$, comparison between NECP-Atlas and NJOY for LEU benchmarks.

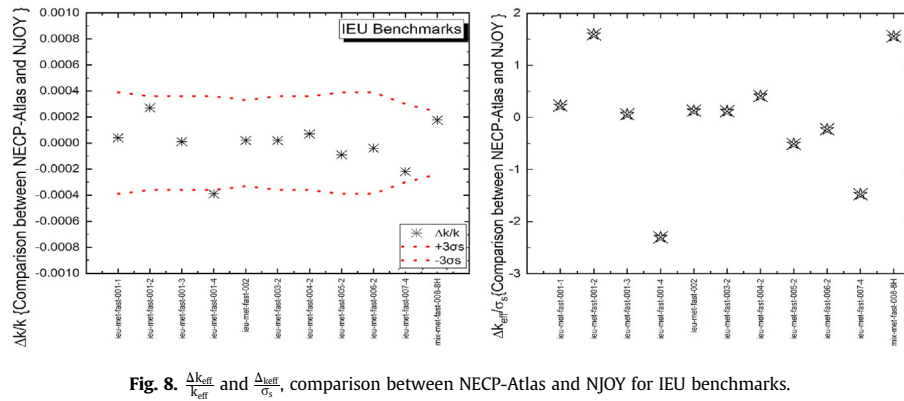


Fig. 8. $\frac{\Delta k_{eff}}{k_{eff}}$ and $\frac{\Delta k_{eff}}{\sigma_s}$, comparison between NECP-Atlas and NJOY for IEU benchmarks.

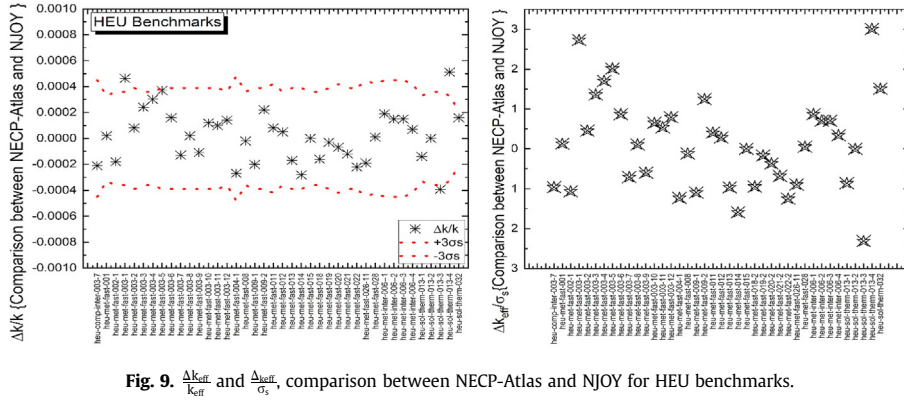


Fig. 9. $\frac{\Delta k_{eff}}{k_{eff}}$ and $\frac{\Delta k_{eff}}{\sigma_s}$, comparison between NECP-Atlas and NJOY for HEU benchmarks.

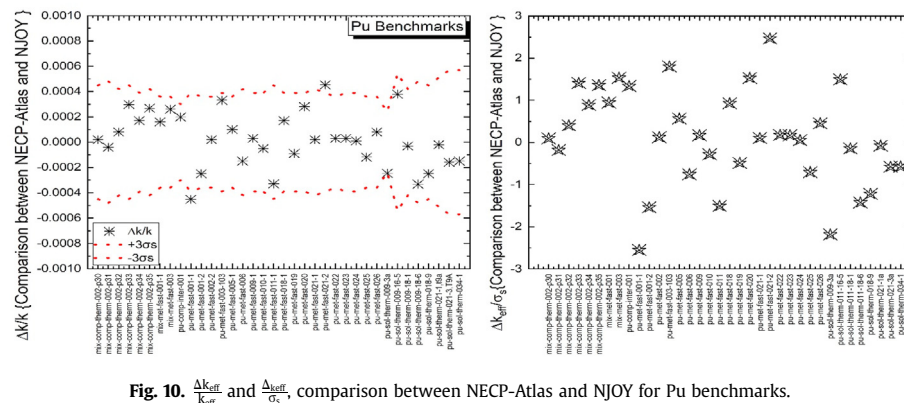


Fig. 10. $\frac{\Delta k_{eff}}{k_{eff}}$ and $\frac{\Delta k_{eff}}{\sigma_s}$, comparison between NECP-Atlas and NJOY for Pu benchmarks.

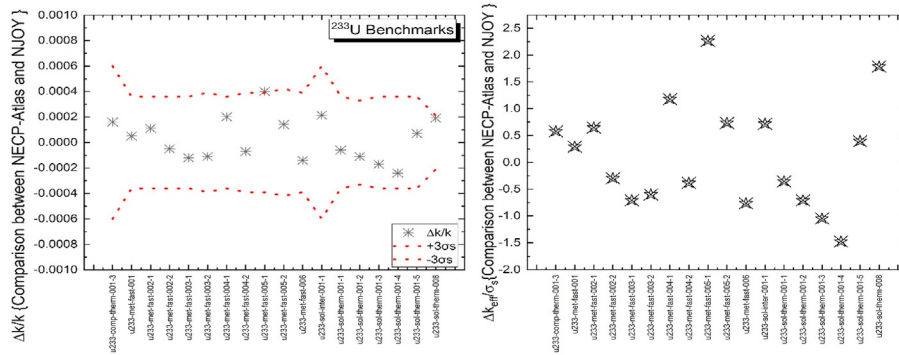


Fig. 11. $\frac{\Delta k_{eff}}{k_{eff}}$ and $\frac{\Delta \rho_{eff}}{\rho_{eff}}$ comparison between NECP-Atlas and NJOY for ^{233}U benchmarks.

Table 3

χ^2 goodness-of-fit, and the $\langle |\Delta| \rangle$ factors, comparison between NECP-Atlas, NJOY and the benchmarks results.

Benchmarks	Number of benchmarks in the category	NECP-Atlas		NJOY	
		χ^2	$\langle \Delta \rangle$	χ^2	$\langle \Delta \rangle$
LEU benchmarks	15	2.34	235.20	2.36	229.53
IEU benchmarks	11	6.71	346.00	6.61	347.55
HEU benchmarks	40	2.58	254.30	2.61	257.63
Pu benchmarks	36	1.44	270.75	1.48	268.86
^{233}U benchmarks	18	1.70	204.39	1.68	204.72
Average	120	2.96	262.13	2.95	261.66

Table 4

Weighted mean $\overline{k_{eff}}$ and S_p .

Benchmarks	Number of benchmarks in the category	NECP-Atlas		NJOY	
		$\overline{k_{eff}} \pm S_p$		$\overline{k_{eff}} \pm S_p$	
LEU benchmarks	15	0.99848	± 0.00292	0.99847	± 0.00299
IEU benchmarks	11	1.00036	± 0.00412	1.00036	± 0.00415
HEU benchmarks	40	0.99980	± 0.00389	0.99977	± 0.00390
Pu benchmarks	36	0.99995	± 0.00371	0.99995	± 0.00373
^{233}U benchmarks	18	0.99939	± 0.00216	0.99937	± 0.00215

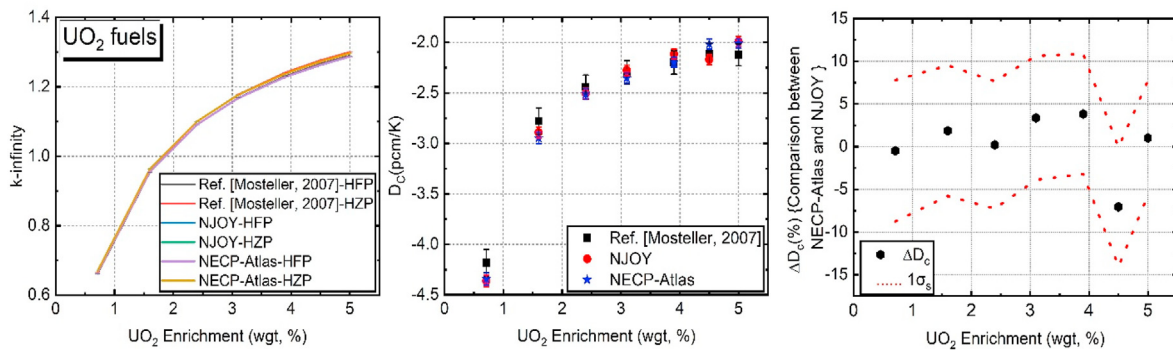


Fig. 12. Doppler Coefficients for UO_2 fuels.

treatments in unresolved resonance range during processing [3]. The maximum observed difference is 7.1% for UO_2 (enriched to 4.5%), but this is generally acceptable since ΔD_C values are mainly (94%) within the $\pm 1\sigma_s$ range near the value given by NJOY.

5. Summary

This work has focused on processing the entire recent release of the ENDF nuclear data library with the NECP-Atlas and NJOY

processing codes. The subject is to perform a statistical inter-comparison between the produced neutron libraries. This study also demonstrates the way ACE libraries are processed for MCNP codes. The verification and the mutual comparison of processed files are carried out by performing criticality calculations for 120 criticality safety benchmarks and additional benchmarks proposed by Russell D. Mosteller.

The results for criticality safety benchmarks are acceptable to good in the vast majority of cases, with only minor differences

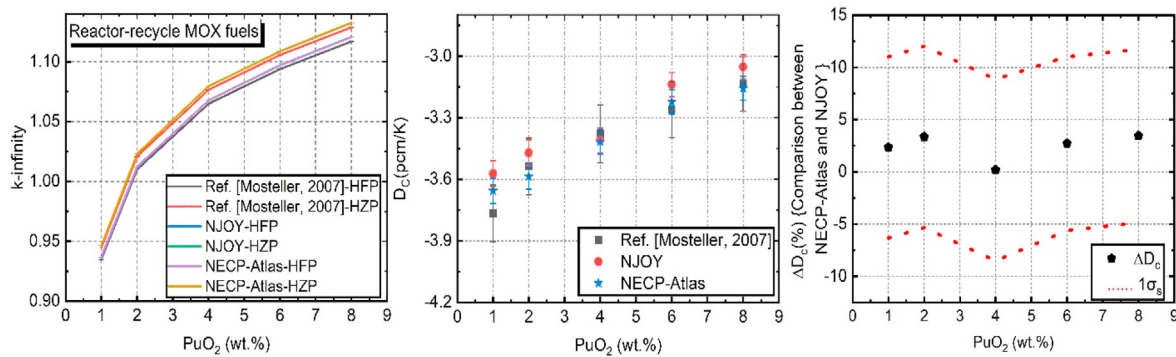


Fig. 13. Doppler Coefficients for Reactor-Recycle MOX fuels.

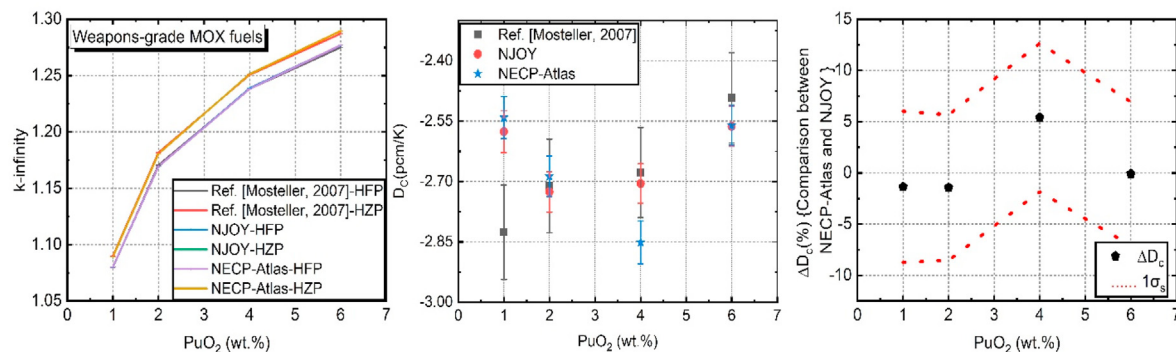


Fig. 14. Doppler Coefficients for Weapons-Grade MOX fuels.

between the two processed libraries and the benchmark values. The largest deviations occurred for some known benchmarks, though such poor performances also have been reported for earlier ENDF/B releases.

For the benchmark experiments used, statistical analyzes based on $\frac{\Delta k_{eff}}{k_{eff}}$, χ^2 , $\langle |\Delta| \rangle$, the weighted mean multiplication factor and the bias uncertainty were established and presented. Results show that the discrepancy between the values obtained for ENDF/B-VIII.0-NECP-Atlas and ENDF/B-VIII.0-NJOY libraries falls within a reasonable range.

The results for the Doppler coefficients are generally good, but in some cases, strong deviations from the benchmark values occur. The largest deviations are seen for Weapons-grade MOX fuels (especially case 1, both libraries). Yet the results showed good agreement with no substantial differences between NECP-Atlas and NJOY libraries.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.net.2021.02.012>.

References

- [1] Csewag, Endf-6 Formats Manual Data, Data formats and procedures for the evaluated nuclear data files ENDF/B-VI, ENDF/B-VII and ENDF/B-VIII, 2018. www.nndc.bnl.gov.
- [2] R. MacFarlane, D.W. Muir, R.M. Boicourt, A.C. Kahler III, J.L. Conlin, The NJOY Nuclear Data Processing System, Version 2016, 2017, <https://doi.org/10.2172/1338791>.
- [3] T. Zu, J. Xu, Y. Tang, H. Bi, F. Zhao, L. Cao, H. Wu, NECP-Atlas: a new nuclear data processing code, Ann. Nucl. Energy 123 (2019) 153–161, <https://doi.org/10.1016/j.anucene.2018.09.016>.
- [4] D.E. Cullen, PREPRO 2019: 2019 ENDF/B pre-processing codes, 2019 report IAEA-NDS-229, <http://www-nds.iaea.org/ndspub/endf/prepro>.
- [5] D. Wiarda, M.E. Dunn, N.M. Greene, M.L. Williams, C. Celik, AMPX-6 : a modular code system for processing ENDF/B. <https://www.ornl.gov/sites/default/files/AMPX-6.pdf>, 2016.
- [6] D.G. Cacuci, Handbook of Nuclear Engineering, vol. 1, Nuclear Engineering Fundamentals, 2010.
- [7] R.E. MacFarlane, A.C. Kahler, Methods for processing ENDF/B-VII with NJOY, Nucl. Data Sheets 111 (2010) 2739–2890, <https://doi.org/10.1016/j.nds.2010.11.001>.
- [8] W.L. Oberkampf, T.G. Trucano, Verification and validation benchmarks, Nucl. Eng. Des. 238 (2008) 716–743, <https://doi.org/10.1016/j.nucengdes.2007.02.032>.
- [9] D.A. Brown, et al., ENDF/B-VIII.0: the 8th major release of the nuclear reaction data library with CIELO-project cross sections, new standards and thermal scattering data, Nucl. Data Sheets 148 (2018) 1–142, <https://doi.org/10.1016/j.nds.2018.02.001>.
- [10] M. Herman, A. Trkov, R. Capote, G.P.A. Nobre, D.A. Brown, R. Arcilla, Y. Danon, A. Plompen, S.F. Mughabghab, Q. Jing, G. Zhigang, L. Tingjin, L. Hanlin, R. Xichao, L. Leal, B.V. Carlson, T. Kawano, M. Sin, S.P. Simakov, K. Guber, Evaluation of neutron reactions on iron isotopes for CIELO and ENDF/B-VIII.0, Nucl. Data Sheets 148 (2018) 214–253, <https://doi.org/10.1016/j.nds.2018.02.004>.
- [11] J. Xu, T. Zu, L. Cao, Development and verification of resonance elastic

- scattering kernel processing module in nuclear data processing code NECP-Atlas, *Prog. Nucl. Energy* 110 (2019) 301–310, <https://doi.org/10.1016/j.pnucene.2018.10.006>.
- [12] T. Zu, J. Xu, L. Cao, A processing method of generating $S(\alpha, \beta, T)$ tables considering resonance elastic scattering kernel for the Monte Carlo codes, *Prog. Nucl. Energy* 122 (2020), <https://doi.org/10.1016/j.pnucene.2020.103262>.
 - [13] Nuclear Energy Agency, in: *International Handbook of Evaluated Criticality Safety Benchmark Experiments*, 2010th ed., 2010.
 - [14] R.D. Mosteller, L.D. Eisenhart, R.C. Little, W.J. Eich, J. Chao, Benchmark calculations for the Doppler coefficient of reactivity, *Nucl. Sci. Eng.* 107 (1991) 265–271, <https://doi.org/10.13182/NSE91-A23789>.
 - [15] R.D. Mosteller, ENDF/B-V, ENDF/B-VI, and ENDF/B-VII.0 results for the Doppler-defect benchmark, *Jt. Int. Top. Meet. Math. Comput. Supercomput. Nucl. Appl. M C + SNA* (2007) 1–8, 2007, <https://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.205.265&rep=rep1&type=pdf>.
 - [16] R.D. Mosteller, The Doppler-defect benchmark: overview and summary of results, *Jt. Int. Top. Meet. Math. Comput. Supercomput. Nucl. Appl. M C + SNA* (2007) 1–13, 2007.
 - [17] D.B. Pelowitz, *MCNPX: User's Manual*, v2. 6.0, 2008.
 - [18] International Atomic Energy Agency - Nuclear Data Section, Index of/public/download-endf, 2018. <https://www-nds.iaea.org/public/download-endf/>, accessed January 12, 2019.
 - [19] Los Alamos National Security, Njoy, Los Alamos Natl. Lab., 2016. <http://www.njoy21.io/>, accessed February 28, 2017.
 - [20] X-5 Monte Carlo Team, *MCNP — A General Monte Carlo N-Particle Transport Code*, 2003. Version 5.
 - [21] W.J. Marshall, B.T. Rearden, *Criticality Safety Validation of SCALE 6.1 with ENDF/B-VII.0 Libraries- ORNL/TM-2011/450*, 2013.
 - [22] O. Kabach, A. Chetaine, A. Jalil, A. Darif, A. Saidi, Processing and benchmarking of evaluated nuclear data file/b-viii.0 β 4 cross-section library by analysis of a series of critical experimental benchmark using the Monte Carlo code MCNP(X) and NJOY2016, *Nucl. Eng. Technol.* 49 (2017) 1610–1616, <https://doi.org/10.1016/j.net.2017.08.017>.
 - [23] J.-C. Jaboulay, P.-Y. Cayla, C. Fausser, F. Damian, Y.-K. Lee, A.L. Puma, J.-C. Trama, TRIPOLI-4® Monte Carlo code ITER A-lite neutronic model validation, *Fusion Eng. Des.* 89 (2014) 2174–2178, <https://doi.org/10.1016/j.fusengdes.2014.02.038>.
 - [24] M. Cornock, Benchmarking of ENDF/B-VII.1 and JENDL-4.0 in the fast neutron range, *Nucl. Data Sheets* 118 (2014) 430–432, <https://doi.org/10.1016/j.nds.2014.04.098>.
 - [25] R.H.H. Smith, J. Keener, J.F. DeClue, A.W. Krass, V.R. Cain, *Bias and Uncertainty of Critical Experiment Models with CSAS25 from SCALE4.4a for Criticality Safety Analyses on the HP J-5600, CMODB Workstation*, 2001.
 - [26] J. Clarity, W.B.J. Marshall, Assessment of normality for criticality safety bias and bias uncertainty calculation, in: *ICNC 2019 - 11th Int. Conf. Nucl. Crit. Saf.* Sept. 15–20, 2019 – Paris, Fr., 2019. <http://energy.gov/downloads/doe-public-access-plan>.
 - [27] Y. Nomura, T. Shimooko, A multiple regression equation for calculated k eff bias errors by criticality code system, *Nucl. Technol.* 65 (1984) 340–349, <https://doi.org/10.13182/NT84-A33416>.
 - [28] R. Mosteller, Comparison of ENDF/B-VII.1 and ENDF/B-VII.0 results for the expanded criticality validation suite for MCNP and for selected additional criticality benchmarks, *Nucl. Data Sheets* 118 (2014) 442–445, <https://doi.org/10.1016/j.nds.2014.04.101>.
 - [29] R.D. Mosteller, S.C. Frankle, P.G. Young, Data testing of ENDF/B-VI with MCNP: critical experiments, thermal-reactor lattices, and time-of-flight measurements, *Adv. Nucl. Sci. Technol.* 24 (2005) 131–195, https://doi.org/10.1007/0-306-47811-0_4.
 - [30] R.D. Mosteller, F.B. Brown, B.C. Kiedrowski, *An expanded criticality validation suite for MCNP*, Tech. Rep. LA-UR-10-04170 (2011).
 - [31] A.C. Kahler, Monte Carlo eigenvalue calculations with ENDF/B-VI.8, JEFF-3.0, and JENDL-3.3 cross sections for a selection of international criticality safety benchmark evaluation project handbook benchmarks, *Nucl. Sci. Eng.* 145 (2003) 213–224, <https://doi.org/10.13182/NSE03-A2377>.
 - [32] O. Kabach, A. Chetaine, A. Benchrif, Processing of JEFF-3.3 and ENDF/B-VIII.0 and testing with critical benchmark experiments and TRIGA Mark II research reactor using MCNPX, *Appl. Radiat. Isot.* 150 (2019) 146–156, <https://doi.org/10.1016/j.apradiso.2019.05.015>.